Statistical Report

Determinants of Belgian Household Energy Consumption

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1 Background

The goal of "*Carbon Neutrality*" is gaining increasing momentum in the European political arena amid substantial indicators of an incoming climate change which might have deleterious effects at the societal, economic and, most importantly, environmental level. Household energy consumption plays a pivotal role in overall social sustainability since it accounts for a substantial share of the total energy demand in most developed countries. Within this sector, electricity consumption provides a useful proxy for household energy demand and can be put into relationship to structural characteristics of the buildings as well as the socio-economic profile of their occupants.

Dwelling-related attributes such as floor area, dwelling type, construction period, and Energy Performance Certificate (EPC) labels are consistently identified as important predictors of electricity demand (McLoughlin et al., 2013; Santin et al., 2009). Intuitively, larger dwellings typically require more energy for lighting, heating, cooling, and appliance use, while older buildings are constructed with materials which provides poorer insulation and therefore larger heat dissipation. Nevertheless, these variables interacts with socio-economic factors. Larger households topically host a higher number of occupants, therefore they can share the consumption for heating, refrigerators and air conditioning lowering the *per capita* consumption. Similarly, high EPC label buildings are prerogative of high-income classes which might use a larger number of appliances and indulge into energy-consuming behaviours going to erode the benefits of a properly insulated house in terms of energy savings (Aydin et al., 2017).

Using household electricity consumption as a proxy for energy demand, this cross-sectional study examines its relationship with building characteristics (e.g. floor area, EPC label, construction era) and socio-economic factors (number of permanent residents and annual net household income) collecting data from 2500 households in Belgium. Importantly, the study aims at characterising the interplay between EPC labels and the other factors involved in energy consumptions to provide data-driven tools to decision-makers.

2 Data management

2.1 Data Cleaning

Three observations (rows 66, 36, and 135) were removed from the dataset according to the instructions. Subsequently, the dataset was cleaned by excluding observations with missing values in the variables EPC label (`epc_label`) and annual household electricity consumption (`annual_kwh`). This step ensures that the analysis is conducted only on complete cases for the key variables of interest. None of the variables displayed missingness above 5%. Missing values in other variables were retained (see appendix) without imputation. To inspect the distribution of electricity consumption and identify potential extreme values, the observations were sorted in decreasing order of annual electricity consumption (`annual_kwh`). Five observations (row 267, 1330, 1570, 369, 12) were two orders of magnitude larger than the others (172 to 312 times larger) and, therefore, were removed as were attributed to imputation error.

2.2 Data Coding

Variables were coded according to their type: continuous, count, ordinal, categorical, or binary. Table 1 summarizes their descriptions and coding schemes.

Table 1: Data Dictionary

Name	Description	Type_Coding
annual_kwh	yearly electricity use	Continuous (kWh/year)
sq_meters	indoor living area	Continuous (m^2)
epc_label	energy performance certificate	Ordinal: A, B, ..., F
building_era	construction period	Ordinal: dummy coded; ref = Pre-1970
income_euro	net household income	Continuous (€/year)
occupancy_count	permanent residents	Count: 1, 2, ..., n
dist_to_brussels	distance to Brussels center	Continuous (km)
roof_color	roof tile color	Categorical: e.g., red, black, other
pet_ownership	owner has a pet	Binary: 0 = No, 1 = Yes

3 Descriptive Statistics

3.1 Categorical variables

The distribution of EPC labels is relatively balanced across categories. Labels A (22.2%) and C (21.3%) are the most common, followed by B (17.6%) and D (15.7%). Lower-efficiency categories E occur less frequently, 13.6% and F 9.6% respectively. Buildings constructed between 1970 and 2000 represent 34.4% of the sample, pre-1970 buildings account for 33.1%, and post-2000 buildings make up 32.6%. Occupancy count ranges from 1 to 7 with all counts accounting approximately 14-15%. Red roofs are the most common (35.2%), followed closely by black roofs (33.3%) and other colours (31.5%). Almost one in four (39.6%) owners has a pet, while 60.4% does not. None of the categories appear too sparse, suggesting that all variables can be retained for modelling without concerns about low cell counts. Overall, the categorical variables show meaningful variation across categories. From the exploratory analysis, two categorical variables seem to correlate strongly with electricity consumption: EPC label and occupancy count. The other categorical variables do njot display an apparent correlation with electricity consumption. Summary tables and plots are presented in the appendix.

3.2 Continuous variables

On average, households consume 5,874.76 kWh of electricity annually, with a relatively large standard deviation (2,224.25 kWh). The median consumption (5,432.07 kWh) is slightly lower than the mean, suggesting a right-skewed distribution, which is also supported by the QQ plot showing upward deviations above the diagonal line in the upper tail (see appendix). Floor area averages 158.25 m^2 with a standard deviation of 36.50 m^2, and a range from 60.26 m^2 to 341.30 m^2. The QQ plot for floor area shows only mild departures from normality. The average distance to Brussels

is 62.96 km, with observations ranging from 5.11 km to 120 km. Its QQ plot indicates noticeable curvature, implying that the distribution deviates from normality, possibly reflecting clustering of households at certain distance ranges during the sampling procedure. Household income averages €48,842.90, with a standard deviation of €22,534.97 and a range from €8,487.91 to €206,367.91. As expected for this type of variable the QQ plot shows strong upper-tail deviation, suggesting positive skewness driven by high-income outliers.

The correlation matrix reveals that annual electricity consumption has a moderate positive correlation with floor area (r = 0.38) and with income (r = 0.29), suggesting that larger and wealthier households tend to consume more energy. Importantly, two predictors exhibit a strong positive correlation with each other, namely floor area and income (r = 0.72). Distance to Brussels shows virtually no correlation with the other variables, suggesting that location relative to Brussels does not strongly influence household energy use, dwelling size, or income in this dataset.

Summary tables and graphs are provided in the appendix.

4 Inferential Statistics

4.1 Regression model

The initial model contained all variables plus interaction terms between income and number of occupants, EPC label and floor area, EPC label and number of occupants, and income and EPC label (see appendix). The outcome variable `annual_kwh` was initially regressed using row data. Diagnostic plots (see appendix) showed some violations of the standard assumptions. In particular, the residuals vs fitted plot indicates the presence of heteroskedasticity, with the variance of the residuals increasing for higher fitted values, and mild deviations from linearity. The outcome variable `annual_kwh` was then log transformed to ensure normal distribution. At each step, the variable whose removal produced the greatest reduction in BIC was eliminated from the model until no further decrease was observed. After applying backward stepwise selection based on the BIC, the final model retained floor area, EPC label, and the number of occupants as predictors of annual electricity consumption. The interaction terms and other covariates were removed. The final model explained approximately 89% of the variability in log-transformed annual electricity consumption. Larger homes, lower EPC ratings, and a higher number of occupants were associated with significantly higher electricity consumption.

Table 2: Regression results for the selected model

	Estimate	Std. Error	t value	p-value
(Intercept)	7.21014	0.01304	552.97203	0
sq_meters	0.00396	0.00007	58.45812	0
epc_labelB	0.08743	0.00787	11.10654	0
epc_labelC	0.15916	0.00748	21.26844	0
epc_labelD	0.24176	0.00814	29.68186	0
epc_labelE	0.32544	0.00850	38.28402	0
epc_labelF	0.40311	0.00953	42.29112	0
occupancy_count	0.15110	0.00125	120.82848	0

4.2 Contrast analysis

- for research question 2 and 4

5 Conclusions

6 Appendix

6.1 Summary tables for non-continuous varibales

Tables below refer to the dataset with only rows 66, 36 135 removed.

Table 3: EPC Label Distribution

epc_label	n	percent
A	554	22.2
B	436	17.5
C	532	21.3
D	391	15.7
E	338	13.5
F	241	9.7
NA	5	0.2

Table 4: Building Era Distribution

building_era	n	percent
1970-2000	856	34.3
Post-2000	814	32.6
Pre-1970	827	33.1

Table 5: Roof Color Distribution

roof_color	n	percent
Black	832	33.3
Other	788	31.6
Red	877	35.1

Table 6: Pet Ownership Distribution

pet_ownership	n	percent
0	1509	60.4
1	988	39.6

Table 7: Occupancy Count Distribution

occupancy_count	n	percent
1	352	14.1
2	357	14.3
3	346	13.9
4	379	15.2
5	377	15.1
6	342	13.7
7	339	13.6
NA	5	0.2

6.2 Summary tables for continuous varibales

Tables below refer to the dataset with only rows 66, 36 135 removed.

Table 8: Summary Statistics for Continuous Variables

Statistic	annual_kwh	sq_meters	dist_to_brussels	income_euro
Mean	14445.58	158.19	63.03	48844.80
SD	194579.20	36.47	33.07	22504.32
Median	5432.60	155.84	63.24	44557.61
IQR	2996.35	46.80	57.70	25871.05
Min	1824.63	60.26	5.11	8487.91
Max	5372190.00	341.30	120.00	206367.91

6.3 Summary plots for continuous varibales

Figures below refer to the dataset used for the regression model from which missing values and outliers in annual energy consumption were removed as explained in section two. From the box plot we observe that several variables exhibit right-skewness and upper-tail outliers, particularly income and energy consumption. While annual energy consumption deviates significantly from normality, its log transformed analogue follows an approximate normal distribution dispute a mild deviation from the expected quantiles in the lower and upper tail.

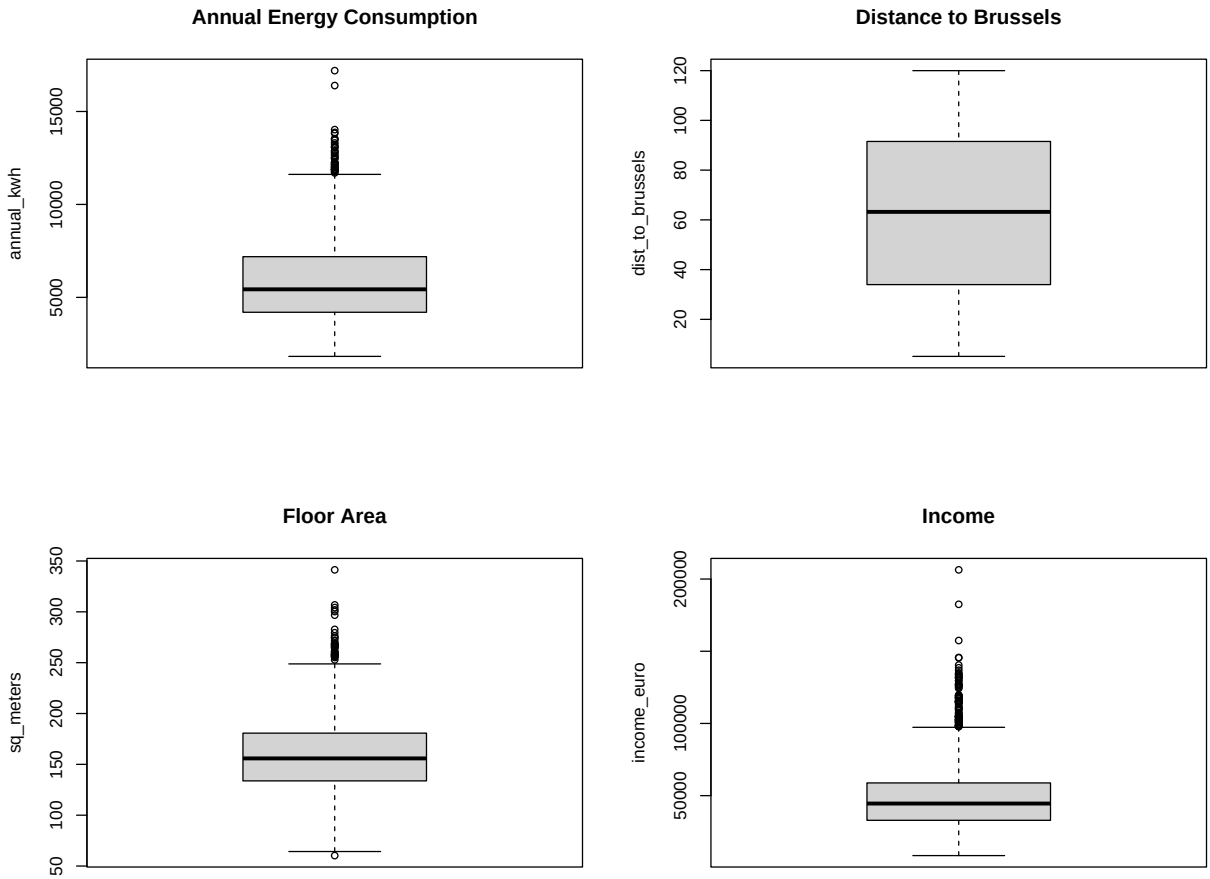


Figure 1: Box plots for continuous variables

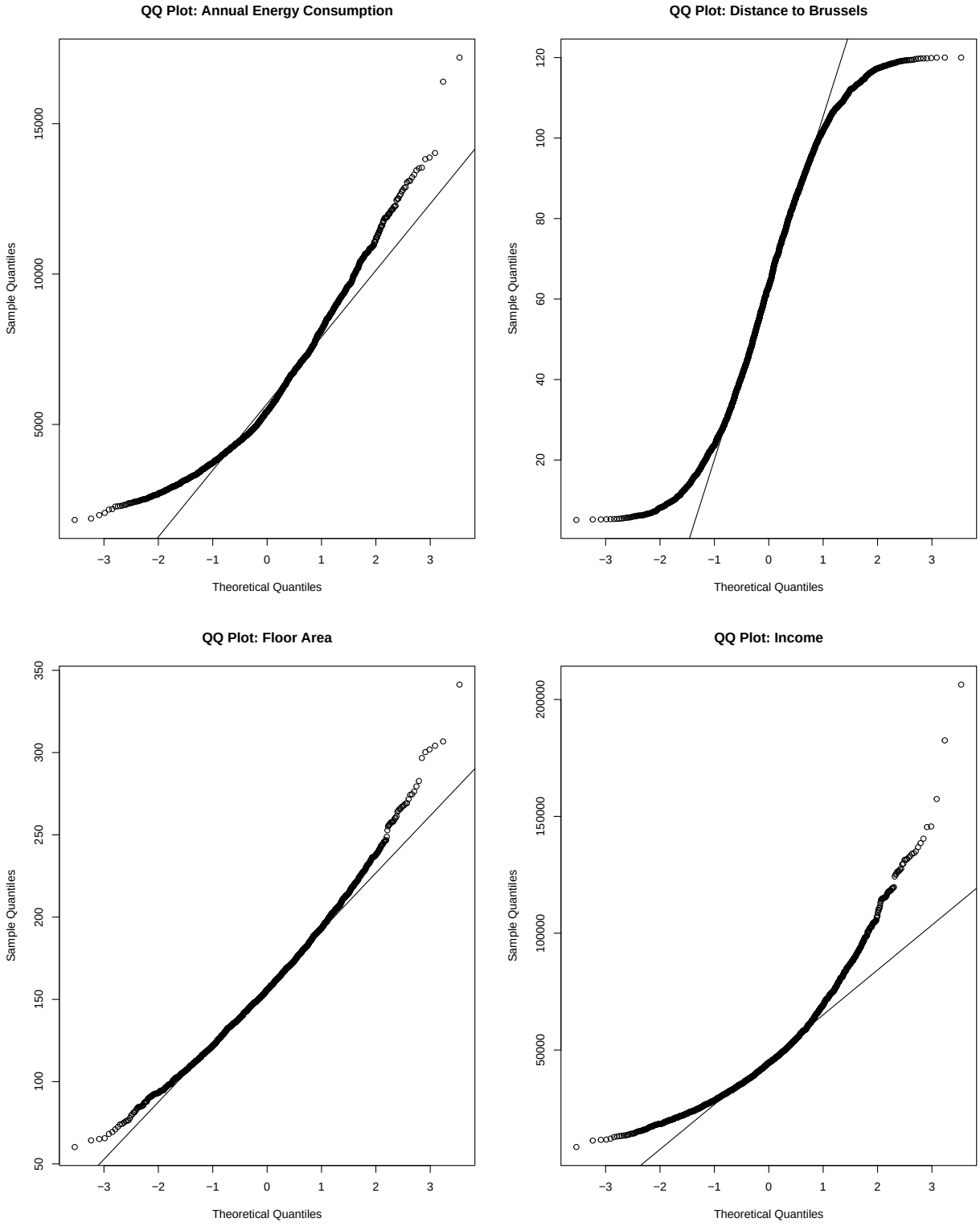


Figure 2: QQ plots for continuous variables

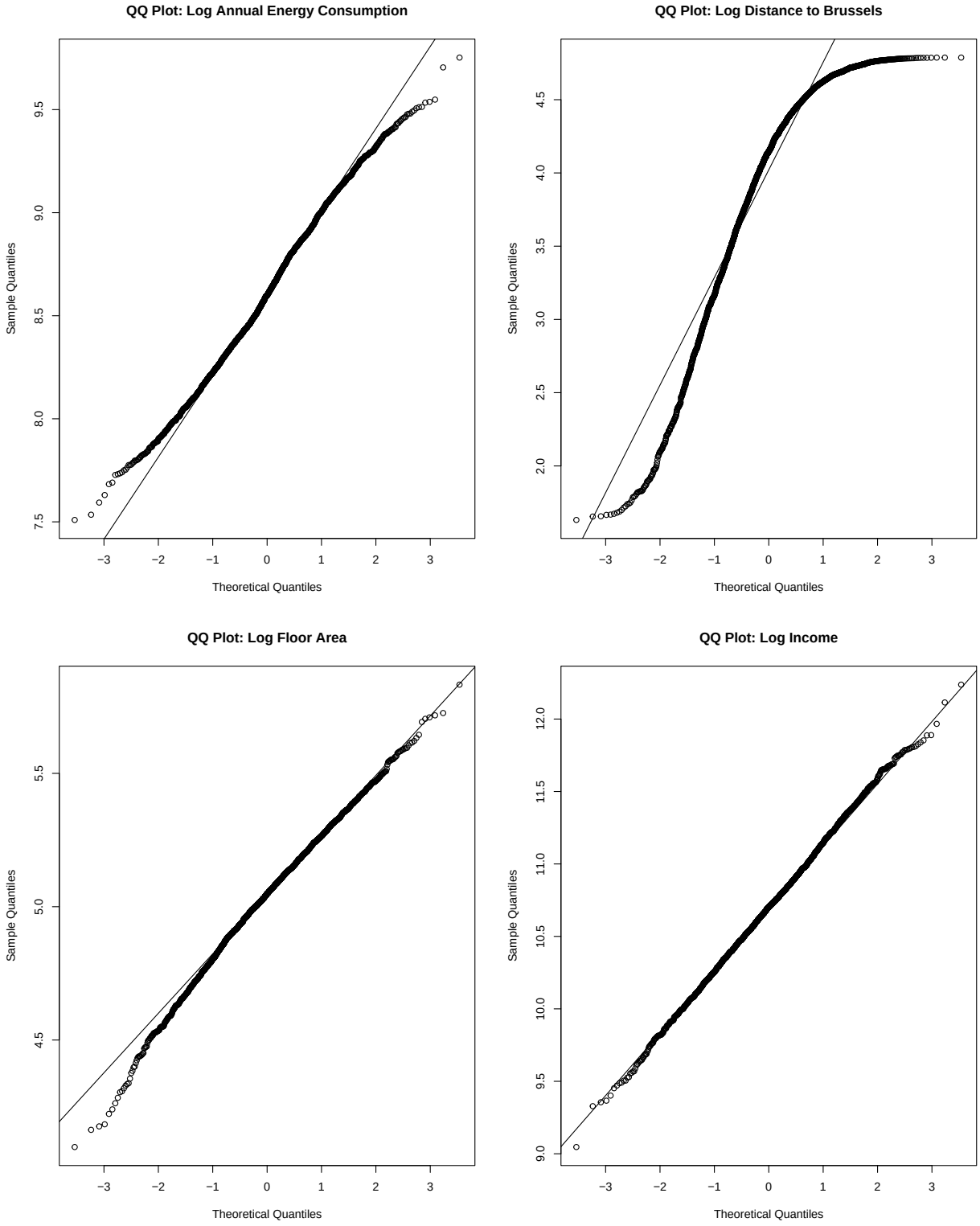


Figure 3: QQ plots for log-transformed continuous variables

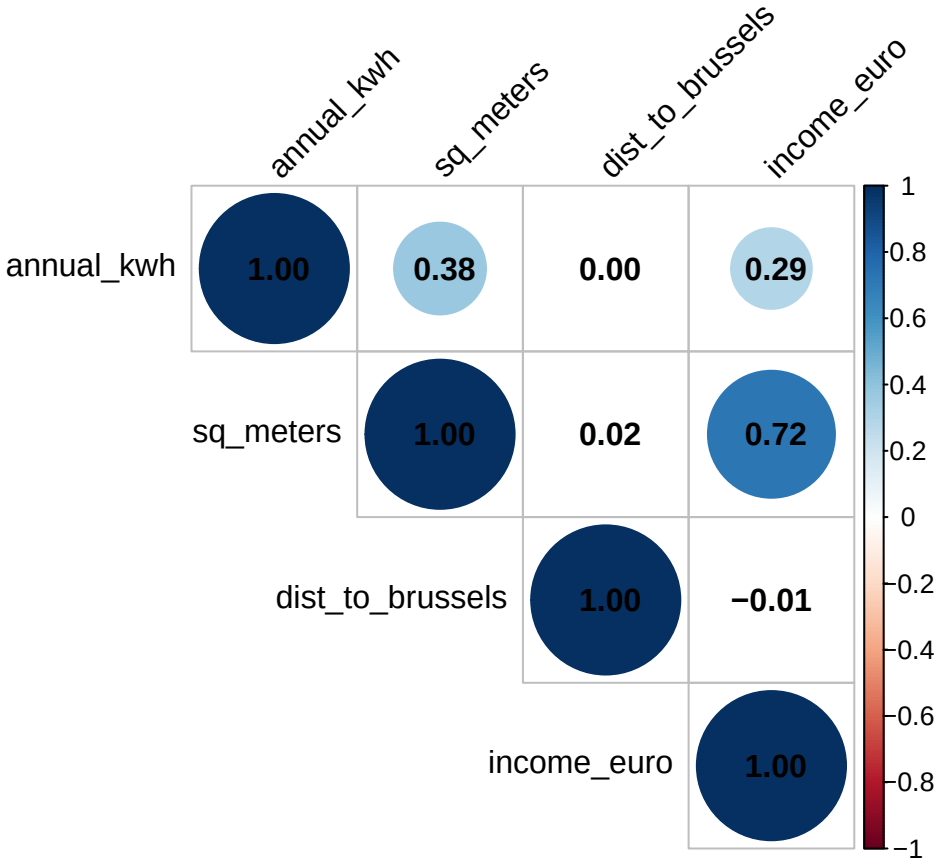
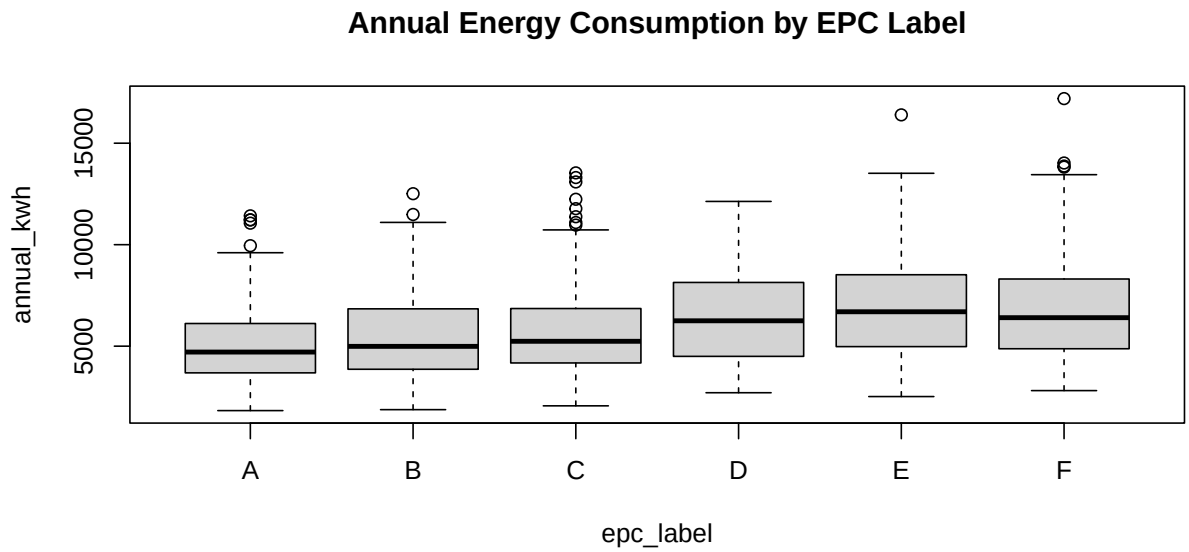


Figure 4: Correlation plots for continuous variables

6.4 Relationship between Annual Energy Consumption and non-continuous variables.

Among categorical and ordinal variables, EPC label and occupancy count are the most correlated to annual energy consumption. In addition, EPC label and building era are correlated with each other since the most efficient labels are clustered in the two most recent building eras. Therefore, building era might play as a confounder in the model.



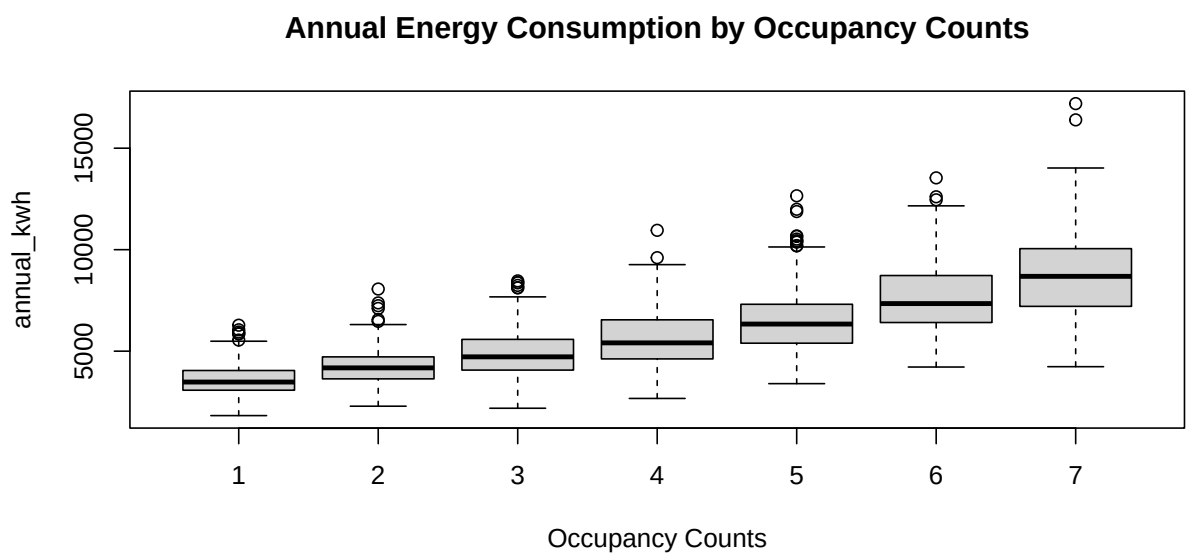


Figure 5: Box plots for annual energy consumption vs categorical variables (part 1)

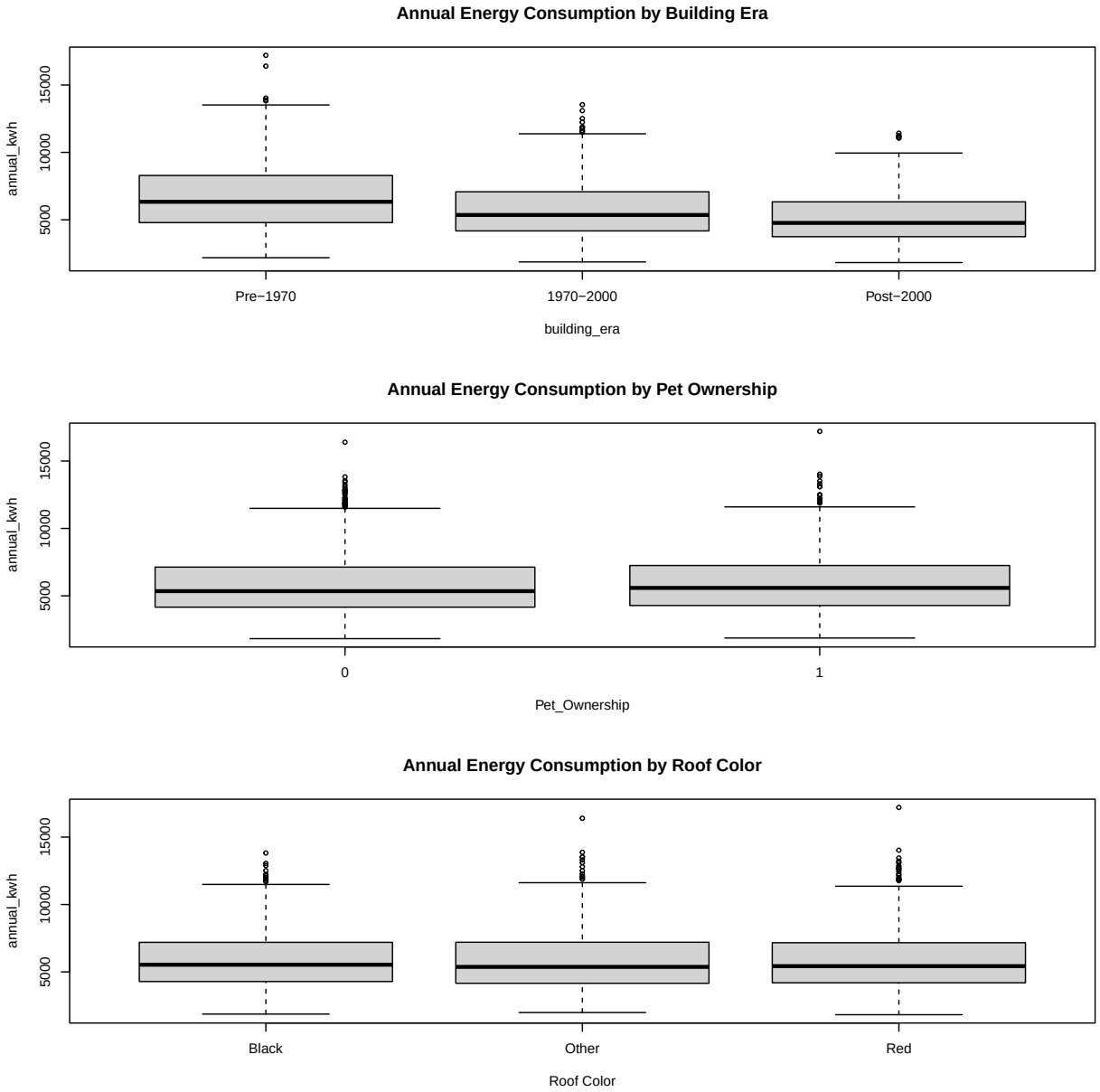


Figure 6: Box plots for annual energy consumption vs categorical variables (part 2)

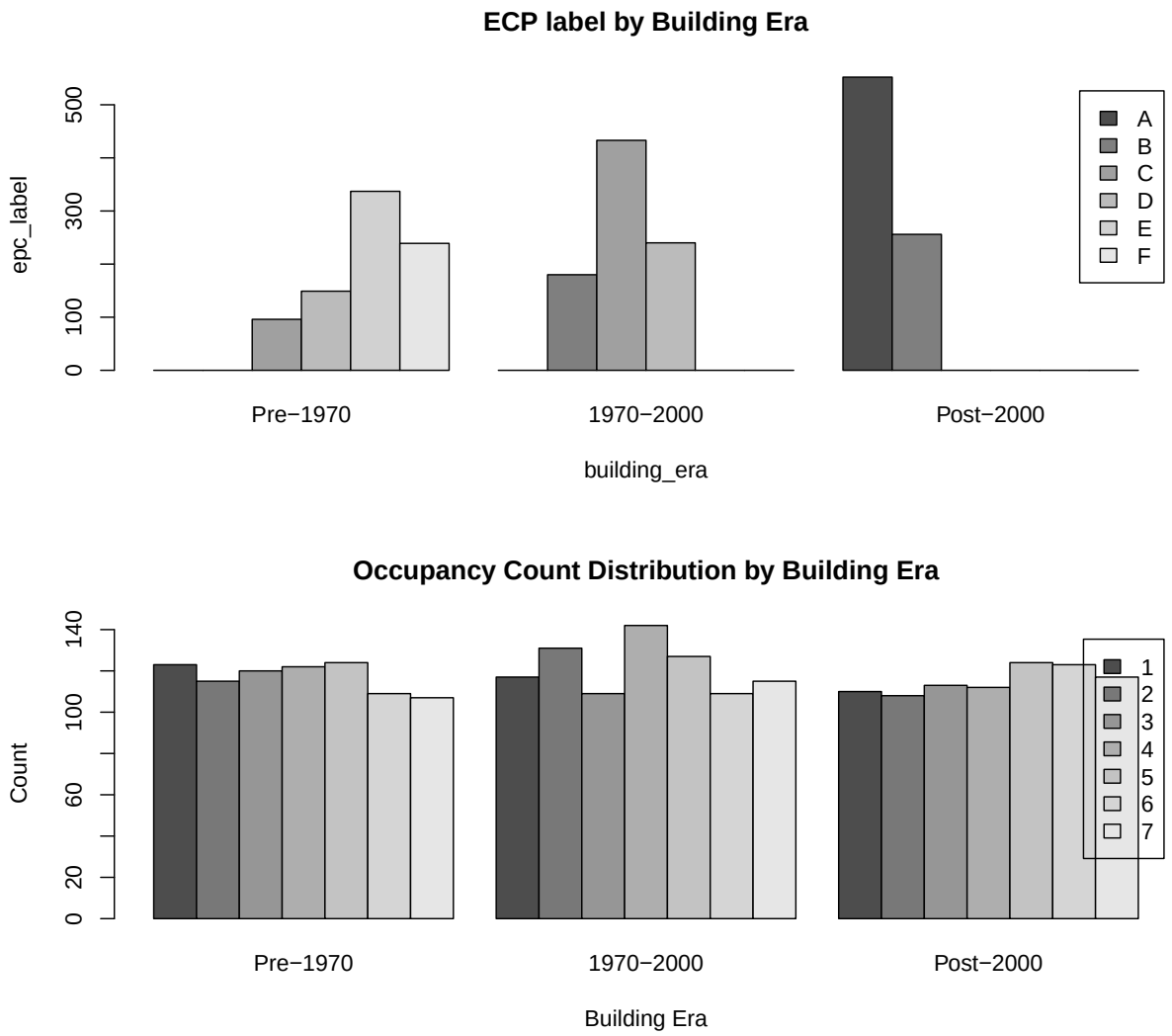


Figure 7: Bar plots predictors against each other (selected cases)

6.5 Regression Model

The initial regression model was the following:

$$
\begin{aligned}
\text{annual_kwh}_i = \beta_0 &+ \beta_1\,\text{sq_meters}_i + \beta_2\,\text{income_euro}_i + \beta_3\,\text{occupancy_count}_i + \beta_4\,\text{dist_to_brussels}_i \\
&+ \boldsymbol{\beta}_5\,\text{epc_label}_i + \boldsymbol{\beta}_6\,\text{building_era}_i + \boldsymbol{\beta}_7\,\text{roof_color}_i + \beta_8\,\text{pet_ownership}_i \\
&+ \beta_9(\text{income_euro}_i \times \text{occupancy_count}_i) \\
&+ \boldsymbol{\beta}_{10}(\text{epc_label}_i \times \text{sq_meters}_i) \\
&+ \boldsymbol{\beta}_{11}(\text{epc_label}_i \times \text{occupancy_count}_i) \\
&+ \boldsymbol{\beta}_{12}(\text{epc_label}_i \times \text{income_euro}_i) + \varepsilon_i
\end{aligned}
$$

The final model was:

$$

$$

6.5.1 Diagnostic plots: Model regressing row annual energy consumption

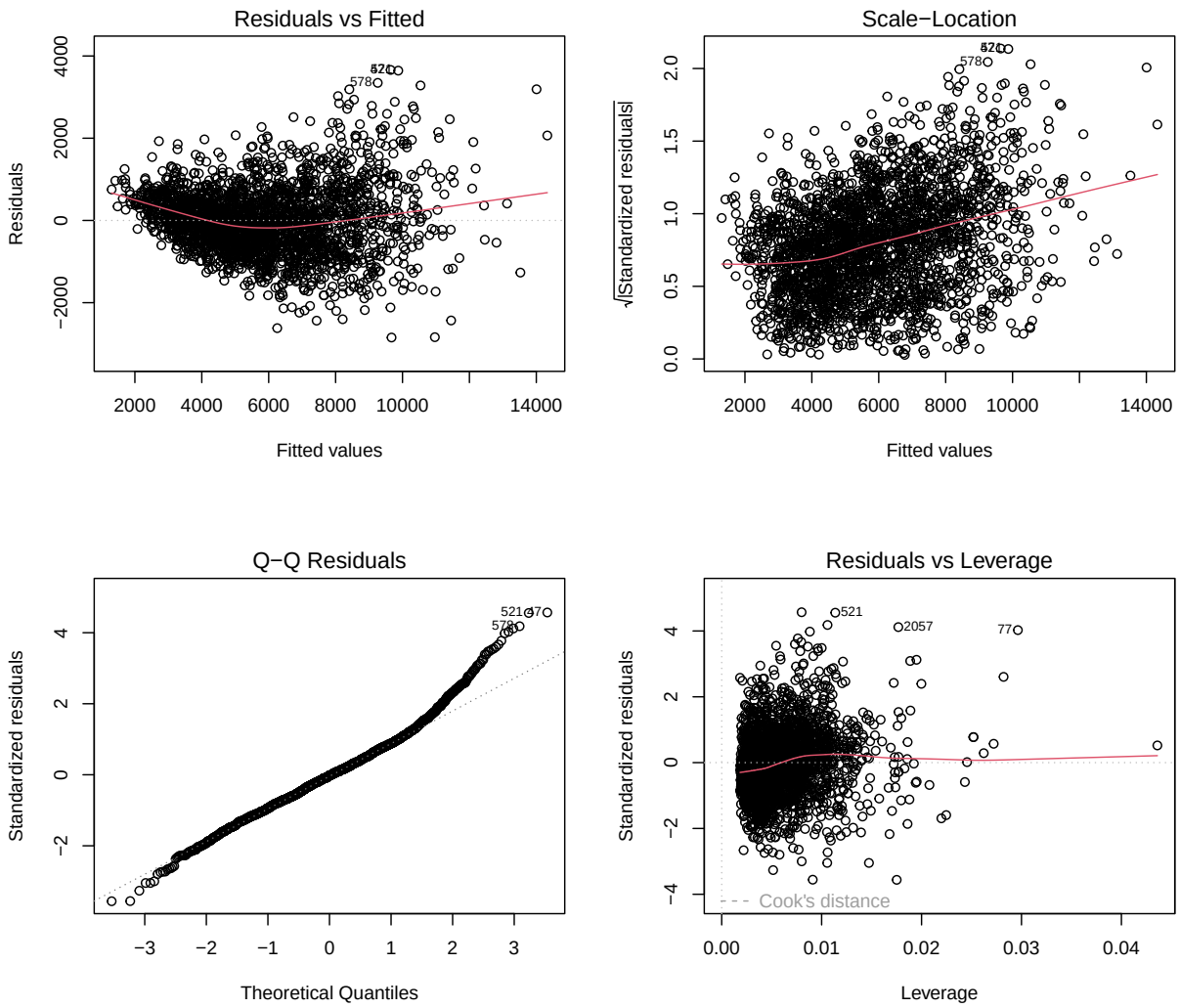


Figure 8: Diagnostic plots for the selected regression model using annual energy consuption without log transformation.

6.5.2 Diagnostic plots: Model regressing log-transformed annual energy consumption

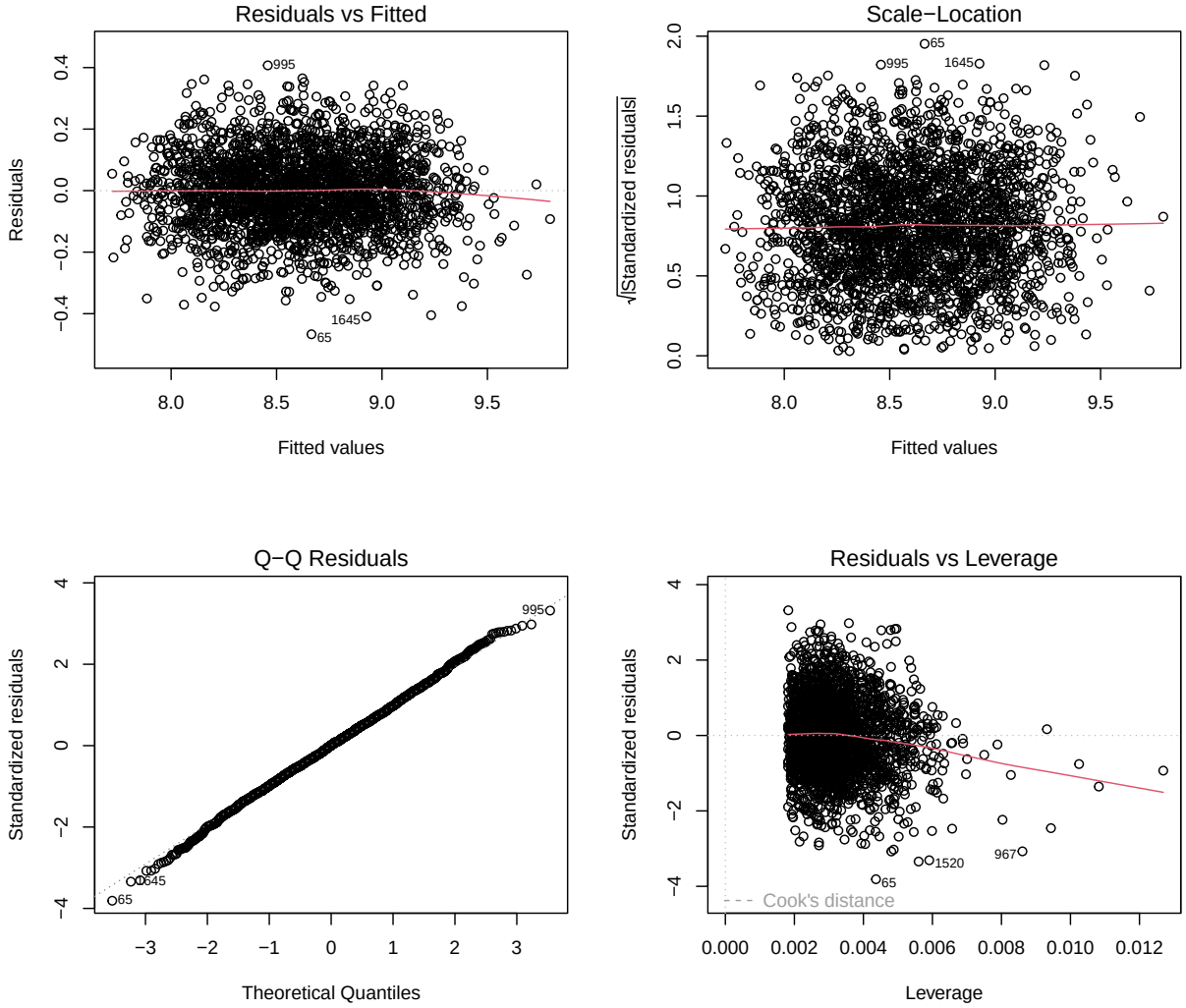


Figure 9: Diagnostic plots for the selected regression model using log-transformed annual energy consuption.

References

Aydin, E., Brounen, D., & Kok, N. (2017). The rebound effect in residential energy consumption. *The RAND Journal of Economics*, *48*(3), 749–782.

McLoughlin, F., Duffy, A., & Conlon, M. (2013). Determinants of residential electricity consumption: Using smart meter data to examine the effect of climate, building characteristics, appliance stock, and occupants' behavior. *Energy*, *55*, 184–194.

Santin, O. G., Itard, L., & Visscher, H. (2009). The effect of occupancy and building characteristics on energy use for space and water heating in dutch residential stock. *Energy and Buildings*, *41*(11), 1223–1232.